CATEGORICAL COPRODUCTS

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1. Introduction

The problem involves the explicit construction of categorical structures that provide a step towards a construction of 4-dimensional topological quantum field theories. The idea of obtaining invariants of smooth 4-dimensional manifolds by constructing field theories extended to dimension 1 by "categorifying" Reshetikhin-Turaev's construction in $3d$ was advocated by Crane and Frenkel [CrFr]. Representations of quantum groups on $\mathbb{C}(q)$-vector spaces are replaced by "2-representations" on categories [ChRou, Lau, KhoLau, Rou1]. The 2-dimensional pair of pants should correspond to a tensor structure on the category of 2-representations. That tensor structure does not arise from a coproduct but from a trimodule and the problem is devoted to the construction of that trimodule. The construction applies to the monoidal category associated to the positive part of a Kac-Moody algebra. Cf [Rou2] for more details.

The first step of the proof (Lemma 2.2) is to show that such a quantization is endowed with non-functorial symmetry maps τ that satisfy braid relations up to terms of low degree. This is where the non-degeneracy assumption (iv) is used.

The construction of the action of $\Delta(c_0)$ on objects is suggested by the statement of the problem. The construction of $\Delta(c_0)(\alpha_c)$ involves the maps τ.

Lemma 2.3 is the crucial tool to establish functoriality. It controls the lack of functoriality of the maps τ.

2. Preliminaries

Note that we have $\mathcal{D}^{\leq 0} = (\operatorname{gr}\mathcal{D})^0 \underset{\sim}{\overset{\Phi}{\to}} (\operatorname{gr}\mathcal{C})^0 = \mathcal{C}^{\leq 0}$.

Given $d, d' \in \mathcal{D}_1$, let $\bar{\sigma}_{d,d'}$ be the image of $\sigma_{d,d'}$ in $\operatorname{Hom}^1_{\operatorname{gr}\mathcal{D}}(dd', d'd)$.

We denote by $\mathcal{C}_1$ the full subcategory of $\mathcal{C}$ with objects those of $\Phi(\mathcal{D}_1)$.

Given $c, c' \in \mathcal{C}_1$, let $\tau_{c,c'} \in \operatorname{Hom}^{\leq 1}_{\mathcal{C}}(cc', c'c)$ be a map whose image in $\operatorname{Hom}^1_{\operatorname{gr}\mathcal{C}}(cc', c'c)$ is mapped to $\bar{\sigma}_{c,c'}$ by Φ. Given $c_1, \dots, c_n \in \mathcal{C}_1$ and $i \in \{1, \dots, n-1\}$, we put $\tau_i = c_1 \cdots c_{i-1} \sigma_{c_i, c_{i+1}} c_{i+2} \cdots c_n$. We fix a reduced decomposition $w = s_{i_1} \cdots s_{i_r}$ for each element $w \in \mathfrak{S}_n$ and we put $\tau_w = \tau_{i_1} \cdots \tau_{i_r}$.

Properties (ii) and (iii) hold for $\mathcal{D}$ replaced by $\operatorname{gr}\mathcal{D}$, hence for $\mathcal{C}$. In (iii), σ_w is replaced by τ_w.

Let $\mathcal{C}'$ be the free strict monoidal category generated by $\mathcal{C}_1$ and maps $\tilde{\tau}_{c,c'} : cc' \to c'c$ for $c, c' \in \mathcal{C}_1$. Let $\Psi : \mathcal{C}' \to \mathcal{C}$ be the strict monoidal functor that is the identity on $\mathcal{C}_1$ and sends $\tilde{\tau}_{c,c'}$ to $\tau_{c,c'}$. Define $\tilde{\mathcal{C}} = \mathcal{C}'/\ker\Psi$. The functor Ψ factors through an equivalence $\tilde{\mathcal{C}} \to \mathcal{C}$. We can replace the category $\mathcal{C}$ by the equivalent category $\tilde{\mathcal{C}}$ to solve the problem. So, we can assume that tensor product is a bijection $\coprod_{n\geq 0} \operatorname{Obj}(\mathcal{C}_1)^n \xrightarrow{\sim} \operatorname{Obj}(\mathcal{C})$. We denote by $\mathcal{C}_n$ the full

subcategory of $\mathcal{C}$ with objects $c_1 \cdots c_n$ with $c_i \in \mathcal{C}_1$ and we put $\mathcal{C}_+ = \coprod_{n\geq 1} \mathcal{C}_n$. Note that we have a category decomposition $\mathcal{C} = \coprod_{n\geq 0} \mathcal{C}_n$.

Given $c_1, \dots, c_{m+n} \in \mathcal{C}_1$, we put $\tau_{c_1\cdots c_m, c_{m+1}\cdots c_{m+n}} = \tau_{(1\to n+1,\dots,m\to m+n, m+1\to 1,\dots,m+n\to n)}$.

Let $c, c', d, d' \in \mathcal{C}$. We define a map

$$h : \mathrm{Hom}_{\mathcal{C}}(c,d) \otimes \mathrm{Hom}_{\mathcal{C}}(c',d') \to \mathrm{Hom}_{\mathcal{C}}(cc', d'd),\ f\otimes f' \mapsto \tau_{d,d'} \circ (ff') - (f'f) \circ \tau_{c,c'}.$$

The next lemma is immediate.

Lemma 2.1. *We have* $h((f'\circ f)\otimes(g'\circ g)) = (g'f')\circ h(f\otimes g) + h(f'\otimes g')\circ(fg)$.

We have

$$h(ff'\otimes g) = \tau_{d_1,d_2} d_1' \circ f h(f'\otimes g) + h(f\otimes g) f' \circ c_1 \tau_{c_1',c_2}$$

for $f : c_1 \to d_1$, $f' : c_1' \to d_1'$ *and* $g : c_2 \to d_2$ *and*

$$h(f\otimes gg') = d_2 \tau_{d_1,d_2'} \circ h(f\otimes g) g' + g h(f\otimes g') \circ \tau_{c_1,c_2} c_2'$$

for $f : c_1 \to d_1$, $g : c_2 \to d_2$ *and* $g' : c_2' \to d_2'$.

Lemma 2.2. *The monoidal category* $\mathcal{C}$*, its full subcategory* $\mathcal{C}_1$ *and its maps* $\tau_{c,c'}$ *satisfy the following properties:*

- $\tau_{\mathbf{1},c} = \tau_{c,\mathbf{1}} = \mathrm{id}_c$ *for all* $c \in \mathcal{C}$
- $h(f\otimes g)$ *is in the image of the tensor product map* $\mathrm{Hom}_{\mathcal{C}}(c,d')\otimes\mathrm{Hom}_{\mathcal{C}}(c',d) \to \mathrm{Hom}_{\mathcal{C}}(cc', d'd)$*, when* $f, g \in \mathcal{C}_1$.
- $\tau_{c_1c_2,c'} = (\tau_{c_1,c'}c_2)\circ(c_1\tau_{c_2,c'})$ *and* $\tau_{c,c_1'c_2'} = (c_1'\tau_{c,c_2'})\circ(\tau_{c,c_1'}c_2')$ *for any* $c, c', c_1, c_2, c_1', c_2' \in \mathcal{C}$.
- $(c_3\tau_{c_1,c_2})\circ(\tau_{c_1,c_3}c_2)\circ(c_1\tau_{c_2,c_3}) - (\tau_{c_2,c_3}c_1)\circ(c_2\tau_{c_1,c_3})\circ(\tau_{c_1,c_2}c_3) \in \mathrm{Hom}_{\mathcal{C}}^{\leq 1}(c_1c_2c_3, c_3c_2c_1)$.

Proof. The first and third properties are clear. The second property follows from the fact that $h(f\otimes g) \in \mathrm{Hom}^{\leq 0}$.

Let us now prove the last property. Let $u_i \in \mathrm{Hom}_{\mathcal{C}_1}(c_i, c_i')$ for $i \in \{1,2,3\}$.

We have $c_3\sigma_{c_1,c_2}\circ\sigma_{c_1,c_3}c_2\circ c_1\sigma_{c_2,c_3} = \sigma_{c_2,c_3}c_1\circ c_2\sigma_{c_1,c_3}\circ\sigma_{c_1,c_2}c_3$, hence

$$c_3\bar\sigma_{c_1,c_2}\circ\bar\sigma_{c_1,c_3}c_2\circ c_1\bar\sigma_{c_2,c_3} = \bar\sigma_{c_2,c_3}c_1\circ c_2\bar\sigma_{c_1,c_3}\circ\bar\sigma_{c_1,c_2}c_3.$$

We deduce that $\delta = c_3\tau_{c_1,c_2}\circ\tau_{c_1,c_3}c_2\circ c_1\tau_{c_2,c_3} - \tau_{c_2,c_3}c_1\circ c_2\tau_{c_1,c_3}\circ\tau_{c_1,c_2}c_3 \in \mathrm{Hom}_{\mathcal{C}}^{\leq 2}(c_1c_2c_3, c_3c_2c_1)$. So there are $\gamma_1 \in \mathrm{Hom}_{\mathcal{C}}^{\leq 0}(c_3c_1c_2, c_3c_2c_1)$ and $\gamma_2 \in \mathrm{Hom}_{\mathcal{C}}^{\leq 0}(c_2c_3c_1, c_3c_2c_1)$ such that

$$\delta = \gamma_1\circ\tau_{c_1,c_3}c_2\circ c_1\tau_{c_2,c_3} + \gamma_2\circ c_2\tau_{c_1,c_3}\circ\tau_{c_1,c_2}c_3 \mod \mathrm{Hom}^{\leq 1}.$$

We define similarly γ_1' and γ_2' by replacing c_i by c_i'.

We have

$$u_3u_2u_1\circ c_3\tau_{c_1,c_2}\circ\tau_{c_1,c_3}c_2\circ c_1\tau_{c_2,c_3} = (-u_3h(u_1\otimes u_2) + c_3'\tau_{c_1',c_2'}\circ u_3u_1u_2)\circ\tau_{c_1,c_3}c_2\circ c_1\tau_{c_2,c_3}$$

$$= -\tau_{c_2',c_3'}c_1'\circ c_2'\tau_{c_1',c_3'}\circ h(u_1\otimes u_2)u_3 + c_3'\tau_{c_1',c_2'}\circ(-h(u_1\otimes u_3)u_2 + \tau_{c_1',c_3'}c_2'\circ u_1u_3u_2)\circ c_1\tau_{c_2,c_3} \mod \mathrm{Hom}^{\leq 1}$$

$$= -\tau_{c_2',c_3'}c_1'\circ c_2'\tau_{c_1',c_3'}\circ h(u_1\otimes u_2)u_3 - c_3'\tau_{c_1',c_2'}\circ\tau_{c_1',c_3'}c_2'\circ u_1h(u_2\otimes u_3) + $$
$$+ c_3'\tau_{c_1',c_2'}\circ\tau_{c_1',c_3'}c_2'\circ c_1'\tau_{c_2',c_3'}\circ u_1u_2u_3 \mod \mathrm{Hom}^{\leq 1}$$

because $c_3'\tau_{c_1',c_2'}\circ c_3'\tau_{c_2',c_1'} \in \mathrm{Hom}^{\leq 1}$.

We have

$$u_3u_2u_1\circ\tau_{c_2,c_3}c_1\circ c_2\tau_{c_1,c_3}\circ\tau_{c_1,c_2}c_3 = (-h(u_2\otimes u_3)u_1 + \tau_{c_2',c_3'}c_1'\circ u_2u_3u_1)\circ c_2\tau_{c_1,c_3}\circ\tau_{c_1,c_2}c_3$$

$$= -c_3'\tau_{c_1',c_2'}\circ\tau_{c_1',c_3'}c_2'\circ u_1h(u_2\otimes u_3) + \tau_{c_2',c_3'}c_1'\circ(-u_2h(u_1\otimes u_3) + c_2'\tau_{c_1',c_3'}\circ u_2u_1u_3)\circ\tau_{c_1,c_2}c_3 \mod \mathrm{Hom}^{\leq 1}$$

$$= -c'_3\tau_{c'_1,c'_2}\circ\tau_{c'_1,c'_3}c'_2\circ u_1h(u_2\otimes u_3) - \tau_{c'_2,c'_3}c'_1\circ c'_2\tau_{c'_1,c'_3}\circ h(u_1\otimes u_2)u_3+$$

$$+\tau_{c'_2,c'_3}c'_1\circ c'_2\tau_{c'_1,c'_3}\circ\tau_{c'_1,c'_2}c'_3\circ u_1u_2u_3 \mod \mathrm{Hom}^{\leq 1}$$

So

$$u_3u_2u_1\circ(c_3\tau_{c_1,c_2}\circ\tau_{c_1,c_3}c_2\circ c_1\tau_{c_2,c_3} - \tau_{c_2,c_3}c_1\circ c_2\tau_{c_1,c_3}\circ\tau_{c_1,c_2}c_3) \mod \mathrm{Hom}^{\leq 1}$$

$$= (c'_3\tau_{c'_1,c'_2}\circ\tau_{c'_1,c'_3}c'_2\circ c'_1\tau_{c'_2,c'_3} - \tau_{c'_2,c'_3}c'_1\circ c'_2\tau_{c'_1,c'_3}\circ\tau_{c'_1,c'_2}c'_3)\circ u_1u_2u_3 \mod \mathrm{Hom}^{\leq 1}$$

$$= (\gamma'_1\circ\tau_{c'_1,c'_3}c'_2\circ c'_1\tau_{c'_2,c'_3} + \gamma'_2\circ c'_2\tau_{c'_1,c'_3}\circ\tau_{c'_1,c'_2}c'_3)\circ u_1u_2u_3 \mod \mathrm{Hom}^{\leq 1}$$

$$= (\tau_{c'_2,c'_3}c'_1\circ c'_2\tau_{c'_1,c'_3}\circ s_{23}s_{12}(\gamma'_1) + c'_3\tau_{c'_1,c'_2}\circ\tau_{c'_1,c'_3}c'_2\circ s_{12}s_{23}(\gamma'_2))\circ u_1u_2u_3 \mod \mathrm{Hom}^{\leq 1}$$

On the other hand,

$$u_3u_2u_1\circ(\gamma_1\circ\tau_{c_1,c_3}c_2\circ c_1\tau_{c_2,c_3} + \gamma_2\circ c_2\tau_{c_1,c_3}\circ\tau_{c_1,c_2}c_3) =$$

$$\tau_{c'_2,c'_3}c'_1\circ c'_2\tau_{c'_1,c'_3}\circ u_2u_1u_3\circ s_{23}s_{12}(\gamma_1) + c'_3\tau_{c'_1,c'_2}\circ\tau_{c'_1,c'_3}c'_2\circ u_1u_3u_2\circ s_{12}s_{23}(\gamma_2) \mod \mathrm{Hom}^{\leq 1}$$

We deduce that

$$\tau_{c'_2,c'_3}c'_1\circ c'_2\tau_{c'_1,c'_3}\circ(u_2u_1u_3\circ s_{12}s_{23}(\gamma_1) - s_{12}s_{23}(\gamma'_1)\circ u_1u_2u_3)+$$

$$+c'_3\tau_{c'_1,c'_2}\circ\tau_{c'_1,c'_3}c'_2\circ(u_1u_3u_2\circ s_{23}s_{12}(\gamma_2) - s_{23}s_{12}(\gamma'_2)\circ u_1u_2u_3)\in\mathrm{Hom}^{\leq 1}$$

It follows that

$$\sigma_{c'_2,c'_3}c'_1\circ c'_2\sigma_{c'_1,c'_3}\circ(u_2u_1u_3\circ s_{12}s_{23}(\gamma_1) - s_{12}s_{23}(\gamma'_1)\circ u_1u_2u_3)+$$

$$+c'_3\sigma_{c'_1,c'_2}\circ\sigma_{c'_1,c'_3}c'_2\circ(u_1u_3u_2\circ s_{23}s_{12}(\gamma_2) - s_{23}s_{12}(\gamma'_2)\circ u_1u_2u_3)\in\mathrm{Hom}^{\leq 1},$$

hence

$$(u_3u_2u_1\circ s_{23}s_{12}(\gamma_1) - s_{23}s_{12}(\gamma'_1)u_3u_1u_2)\circ\sigma_{c'_1,c'_3}c'_2\circ c'_1\sigma_{c'_2,c'_3}+$$

$$+(u_3u_2u_1\circ s_{12}s_{23}(\gamma_2) - s_{12}s_{23}(\gamma'_2)u_2u_3u_1)\circ c'_2\sigma_{c'_1,c'_3}\circ\sigma_{c'_1,c'_2}c'_3\in\mathrm{Hom}^{\leq 1}.$$

As a consequence, $u_3u_2u_1\circ s_{23}s_{12}(\gamma_1) = s_{23}s_{12}(\gamma'_1)u_3u_1u_2$ and $u_3u_2u_1\circ s_{12}s_{23}(\gamma_2) = s_{12}s_{23}(\gamma'_2)u_2u_3u_1$. This forces $\gamma_i = \gamma'_i = 0$ for $i\in\{1,2\}$. □

Note that the monoidal category $\mathcal{C}^{\mathrm{opp}}$, together with $\mathcal{D}^{\mathrm{opp}}$, satisfies the same assumptions as $\mathcal{C}$ and $\mathcal{D}$. Given f a map in $\mathcal{C}$, we denote by f^{opp} the same map viewed in $\mathcal{C}^{\mathrm{opp}}$. We choose $(\tau^{\mathrm{opp}}_{d,c})$ as the τ-maps in $\mathcal{C}^{\mathrm{opp}}$. We have $h(f\otimes f')^{\mathrm{opp}} = -h(f'^{\mathrm{opp}}\otimes f^{\mathrm{opp}})$.

Given $c_1,c_2,c'_1,c'_2\in\mathcal{C}$, we define the k-submodule $L(c_1,c_2,c'_1,c'_2)$ of $\mathrm{Hom}_{\mathcal{C}}(c_1c_2,c'_1c'_2)$ as the image of the map

$$\bigoplus_{\substack{a,b\in\mathcal{C}\\ e\in\mathcal{C}_+}}\mathrm{Hom}(c_2,ae)\otimes\mathrm{Hom}(c_1a,c'_1b)\otimes\mathrm{Hom}(be,c'_2)\to\mathrm{Hom}(c_1c_2,c'_1c'_2)$$

$$f\otimes g\otimes h\mapsto c'_1h\circ ge\circ c_1f.$$

Lemma 2.3. *Given $c_1,c_2,c'_1,c'_2\in\mathcal{C}$, $f\in\mathrm{Hom}_{\mathcal{C}}(c_1,c'_1)$ and $g\in\mathrm{Hom}_{\mathcal{C}}(c_2,c'_2)$, we have $h(f\otimes g)\in L(c_1,c_2,c'_2,c'_1)$.*

Proof. We will say that the lemma holds for (n_1, n_2) if it holds for all f and g such that $c_i \in \mathcal{C}_{n_i}$.

Let $L_{a,b,e}(c_1, c_2, c_1', c_2')$ be the image of the map $\lambda = \lambda_{a,b,e}$

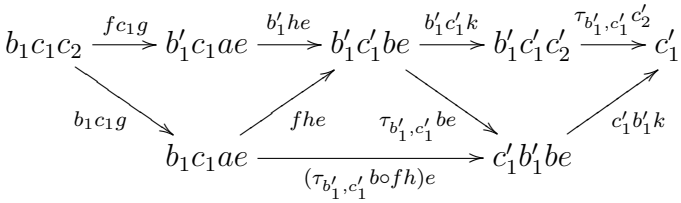
$$\lambda : \mathrm{Hom}(c_2, ae) \otimes \mathrm{Hom}(c_1 a, c_1' b) \otimes \mathrm{Hom}(be, c_2') \to \mathrm{Hom}(c_1 c_2, c_1' c_2'),\ \ f \otimes g \otimes h \mapsto c_1' h \circ ge \circ c_1 f.$$

Note that

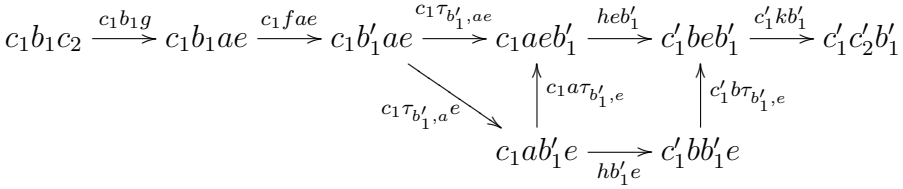
$$L_{a,b,e}(c_1, c_2, c_1', c_2')^{\mathrm{opp}} = L_{b,a,e}(c_1'^{\mathrm{opp}}, c_2'^{\mathrm{opp}}, c_1^{\mathrm{opp}}, c_2^{\mathrm{opp}}).$$

- Here are some properties of those k-submodules of $\mathrm{Hom}(c_1 c_2, c_1' c_2')$.
  - (i) $\mathrm{id}_{c_1 c_2} = \lambda(\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{id}) \in L_{1,1,c_2}(c_1, c_2, c_1, c_2)$ when $c_2 \in \mathcal{C}_+$
  - (ii) $\mathrm{id}_{c_1 d_1 c_2} = \lambda_{1,d_1,c_2}(\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{id}) = \lambda_{d_1,1,c_2}(\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{id})$ is an element of the intersection $L_{1,d_1,c_2}(c_1 d_1, c_2, c_1, d_1 c_2) \cap L_{d_1,1,c_2}(c_1, d_1 c_2, c_1 d_1, c_2)$ when $c_2 \in \mathcal{C}_+$
  - (iii) Given $u_i : b_i \to c_i$ and $v_i : c_i' \to c_i''$ for $i \in \{1, 2\}$ and given $\phi = \lambda(\psi) \in L_{a,b,e}(c_1, c_2, c_1', c_2')$, we have $v_1 v_2 \circ \phi \circ u_1 u_2 = \lambda((\mathrm{id} \otimes v_1 b \otimes v_2) \circ \psi \circ (u_2 \otimes u_1 a \otimes \mathrm{id})) \in L_{a,b,e}(b_1, b_2, c_1'', c_2'')$
  - (iv) $c_1 \tau_{d_1,c_2} = \lambda(\mathrm{id} \otimes \mathrm{id} \otimes \tau_{d_1,c_2}) \in L_{1,d_1,c_2}(c_1 d_1, c_2, c_1, c_2 d_1)$ and $\tau_{c_1,d_1} c_2 = \lambda(\mathrm{id} \otimes \tau_{c_1,d_1} \otimes \mathrm{id}) \in L_{d_1,1,c_2}(c_1, d_1 c_2, d_1 c_1, c_2)$ when $c_2 \in \mathcal{C}_+ 0$
  - (v) Given $f : b_1 \to b_1'$ and $\phi \in L_{a,b,e}(c_1, c_2, c_1', c_2')$, we have $\tau_{b_1',c_1'} c_2' \circ f\phi \in L_{a,b_1' b,e}(b_1 c_1, c_2, c_1', b_1' c_2')$
  - (vi) Consider $f : b_1 \to b_1'$ and $\phi \in L(c_1, c_2, c_1', c_2')$ and assume the lemma holds for (n_1, n_2), where $b_1 \in \mathcal{C}_{n_1}$ and $c_2 \in \mathcal{C}_{n_2}$. We have $\phi f \circ c_1 \tau_{b_1,c_2} \in L(c_1 b_1, c_2, c_1', c_2' b_1')$.

(i), (ii), (iii) and (iv) are clear. (v) follows from the commutativity of the following diagram

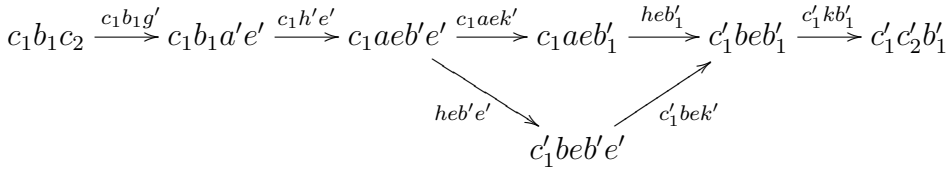


Consider $g : c_2 \to ae$, $h : c_1 a \to c_1' b$ and $k : be \to c_2'$. There is a commutative diagram



It follows that $(c_1' k) \circ he) b_1' \circ c_1(\tau_{b_1',ae} \circ fg) = \lambda(g \otimes (c_1 fa \circ c_1 \tau_{b_1',a} \circ hb_1') \otimes (b\tau_{b_1',e} \circ kb_1')) \in L_{a,bb_1',e}(c_1 b_1, c_2, c_1', c_2' b_1')$.

Consider now $g' : c_2 \to a'e'$, $h' : b_1 a' \to aeb'$ and $k : b'e' \to b_1'$. There is a commutative diagram



It follows that $(c_1' k \circ he) b_1' \circ c_1(aek' \circ h'e' \circ b_1 g') = \lambda(g' \otimes (heb' \circ c_1 h') \otimes kk') \in L_{a',beb',e'}(c_1 b_1, c_2, c_1', c_2' b_1')$.

By assumption, $h(f\otimes g)\in L(b_1,c_2,ae,b_1')$, hence $(c_1'k\circ he)b_1'\circ c_1h(f\otimes g)\in L_{a',be b',e'}(c_1b_1,c_2,c_1',c_2'b_1')$. This shows $(c_1'k\circ he\circ c_1g)f\circ c_1\tau_{b_1,c_2}=(c_1'k\circ he)b_1'\circ c_1(\tau_{b_1',ae}\circ fg-h(f\otimes g))\in L(c_1b_1,c_2,c_1',c_2'b_1')$, hence (vi) holds.

• Let us check the lemma for generating maps of the category $\mathcal{C}$.

Assume d_1, e_1 and c_1' are in $\mathcal{C}_1$. We have

$$h(\tau_{d_1,e_1}\otimes c_1')=\tau_{e_1d_1,c_1'}\circ\tau_{d_1,e_1}c_1'-c_1'\tau_{d_1,e_1}\circ\tau_{d_1e_1,c_1'}$$

$$=\tau_{e_1,c_1'}d_1\circ e_1\tau_{d_1,c_1'}\circ\tau_{d_1,e_1}c_1'-c_1'\tau_{d_1,e_1}\circ\tau_{d_1,c_1'}e_1\circ d_1\tau_{e_1,c_1'}.$$

Lemma 2.2 shows now that $h(\tau_{d_1,e_1}\otimes c_1')$ is a linear combination of maps of the form

- $uvw\in L_{1,1,c_1'}(d_1e_1,c_1',c_1',e_1d_1)$ with $u:d_1\to c_1'$, $v:e_1\to e_1$ and $w:c_1'\to d_1$ (by (ii) and (iii))
- $uvw\circ d_1\tau_{e_1,c_1'}\in L_{1,e_1,c_1'}(d_1e_1,c_1',c_1',e_1d_1)$ with $u:d_1\to c_1'$, $v:c_1'\to e_1$ and $w:e_1\to d_1$ (by (iv) and (iii))
- $uvw\circ\tau_{d_1,e_1}c_1'\in L_{1,e_1,c_1'}(d_1e_1,c_1',c_1',e_1d_1)$ with $u:e_1\to c_1'$, $v:d_1\to e_1$ and $w:c_1'\to d_1$ (by (ii) and (iii)).

We deduce that $h(\tau_{d_1,e_1}\otimes c_1')\in L(d_1e_1,c_1',c_1',e_1d_1)$.

We have

$$h(c_1'\otimes\tau_{d_1,e_1})=e_1\tau_{c_1',d_1}\circ\tau_{c_1',e_1}d_1\circ c_1'\tau_{d_1,e_1}-\tau_{d_1,e_1}c_1'\circ d_1\tau_{c_1',e_1}\circ\tau_{c_1',d_1}e_1.$$

So, $h(c_1'\otimes\tau_{d_1,e_1})$ is a linear combination of maps of the form

- $uvw\in L_{d_1,e_1,1}(c_1',d_1e_1,e_1d_1,c_1')$ with $u:c_1'\to e_1$, $v:d_1\to d_1$ and $w:e_1\to c_1'$ (by (ii) and (iii))
- $uvw\circ c_1'\tau_{d_1,e_1}\in L_{e_1,1,d_1}(c_1',d_1e_1,e_1d_1,c_1')$ with $u:c_1'\to e_1$, $v:d_1\to c_1'$ and $w:e_1\to d_1$ (by (ii) and (iii))
- $uvw\circ\tau_{c_1',d_1}e_1\in L_{d_1,1,e_1}(c_1',d_1e_1,e_1d_1,c_1')$ with $u:c_1'\to d_1$, $v:d_1\to e_1$ and $w:e_1\to c_1'$ (by (iv) and (iii)).

We deduce that $h(c_1'\otimes\tau_{d_1e_1})\in L(c_1',d_1e_1,e_1d_1,c_1')$.

If $c_1,c_1',c_2,c_2'\in\mathcal{C}_1$, then $h(f\otimes g)\in L(c_1,c_2,c_2',c_1')$ by (i) and (iii) and Lemma 2.2.

• Let us now prove stability properties of the statement of the lemma.

Consider $f'\in\mathrm{Hom}(c_1',c_1'')$ and $g'\in\mathrm{Hom}(c_2',c_2'')$. Lemma 2.1 and (iii) show that if $h(f\otimes g)\in L(c_1,c_2,c_2',c_1')$ and $h(f'\otimes g')\in L(c_1',c_2',c_2'',c_1'')$, then $h((f'\circ f)\otimes(g'\circ g))\in L(c_1,c_2,c_2'',c_1'')$.

Consider now $f:c_1\to d_1$, $f':c_1'\to d_1'$ and $g:c_2\to d_2$. We assume $h(f\otimes g)\in L(c_1,c_2,d_2,d_1)$ and we assume that the lemma holds for (n_1,n_2), where $c_1'\in\mathcal{C}_{n_1}$ and $c_2\in\mathcal{C}_{n_2}$. By Lemma 2.1, we have

$$h(ff'\otimes g)=\tau_{d_1,d_2}d_1'\circ fh(f'\otimes g)+h(f\otimes g)f'\circ c_1\tau_{c_1',c_2}.$$

It follows from (iii) and (vi) that $h(f\otimes g)f'\circ c_1\tau_{c_1',c_2}\in L(c_1c_1',c_2,d_2,d_1d_1')$ and from (iii) and (v) that $\tau_{d_1,d_2}d_1'\circ fh(f'\otimes g)\in L(c_1c_1',c_2,d_2,d_1d_1')$. So $h(ff'\otimes g)\in L(c_1c_1',c_2,d_2,d_1d_1')$.

Consider finally $f:c_1\to d_1$, $g:c_2\to d_2$ and $g':c_2'\to d_2'$. Assume $h(f\otimes g')\in L(c_1,c_2',d_2',d_1)$ and assume the lemma holds for (n_1,n_2) where $c_1\in\mathcal{C}_{n_1}$ and $c_2\in\mathcal{C}_{n_2}$. We have

$$h(f\otimes gg')=-h(g^{\mathrm{opp}}g'^{\mathrm{opp}}\otimes f^{\mathrm{opp}})^{\mathrm{opp}}$$

Since $h(g^{\mathrm{opp}} \otimes f^{\mathrm{opp}}) \in L(d_2, d_1, c_1, c_2)$ and $h(g'^{\mathrm{opp}} \otimes f) \in L(d_2', d_1, c_1, c_2')$, the study above applied to $\mathcal{C}^{\mathrm{opp}}$ shows that $h(g'^{\mathrm{opp}} g^{\mathrm{opp}} \otimes f^{\mathrm{opp}}) \in L(d_2 d_2', d_1, c_1, c_2 c_2')$, hence $h(f \otimes gg') \in L(c_1, c_2 c_2', d_2 d_2', d_1)$.

The maps in the monoidal category $\mathcal{C} \otimes \mathcal{C}$ are generated by maps $f \otimes \mathrm{id}$, $\mathrm{id} \otimes f$, $\tau_{d,e} \otimes \mathrm{id}$ and $\mathrm{id} \otimes \tau_{d,e}$ for f a map in $\mathcal{C}_1$ and $d, e \in \mathcal{C}_1$. Since the lemma holds for those maps, we deduce by induction on (n_1, n_2) that the lemma holds in general, using the stability under composition and tensor product. □

Lemma 2.4. *Given $c, c' \in \mathcal{C}$ with $cc' \not\cong 0$, we have $\tau_{c,c'} \notin L(c, c', c', c)$.*

Proof. We have $c = c_1 \cdots c_n$ and $c' = c_1' \cdots c_{n'}'$ for some $c_1, \dots, c_n, c_1', \dots, c_{n'}' \in \mathcal{C}_1$. Consider $v \in \mathfrak{S}_{n'}$, $w \in \mathfrak{S}_{n+r}$ and $x \in \mathfrak{S}_n$ and $f', g', h' \in \mathrm{Hom}^{\leq 0}$. Let $v' \in \mathfrak{S}_{n+n'}$ (resp. w', resp. x') be defined by $v'(i) = i$ for $1 \leq i \leq n$ and $v'(n+i) = n + v(i)$ for $1 \leq i \leq n'$ (resp. $w'(i) = w(i)$ for $1 \leq i \leq n+r$, $w'(n+r+i) = n+r+i$ for $1 \leq i \leq n'-r$, resp. $x'(i) = i$ for $1 \leq i \leq n'$ and $x'(n'+i) = n' + x(i)$ for $1 \leq i \leq n$). We have $x'w'v'(n + v^{-1}(n')) = n' + x(n) > n'$, hence the permutation $x'w'v'$ is not equal to the permutation $p = (1 \to n'+1, \dots, n \mapsto n'+n, n+1 \mapsto 1, \dots, n'+n \mapsto n')$.

Consider $r \in \{1, \dots, n'-1\}$, $a \in \mathcal{C}_r$, $e \in \mathcal{C}_{n'-r}$, $b \in \mathcal{C}_{n+r-n'}$, $f = \tau_v \circ f'$, $g = \tau_w \circ g'$ and $h = \tau_x \circ h'$ with $f', g', h' \in \mathrm{Hom}^{\leq 0}$. The element $\lambda_{a,b,e}(f \otimes g \otimes h)$ is a linear combination of terms of the form $\tau_{x'} \circ \tau_{w'} \circ \tau_{v'} \circ k$ where $k \in \mathrm{Hom}^{\leq 0}$, $x'(i) = i$ for $1 \leq i \leq n'$, $w'(n+r+i) = n+r+i$ for $1 \leq i \leq n'-r$ and $v'(i) = i$ for $1 \leq i \leq n$. So $L_{a,b,e}(c, c', c', c) \subset \bigoplus_{w \in \mathfrak{S}_{n+n'}, w \neq p} (\tau_w \circ \mathrm{Hom}^{\leq 0})$ and the lemma follows. □

3. Functor on objects

We define a functor $\delta : \mathcal{C} \to \overline{\delta\mathcal{C}}$. We put

$$\delta(c) = \mathrm{coker}\Big(\sum_{\substack{d_1 \in \mathcal{C}, d_2 \in \mathcal{C}_+ \\ f \in \mathrm{Hom}_{\mathcal{C}}(d_1 d_2, c)}} (f*) \circ (d_1 \alpha_{d_2}) : \bigoplus_{\substack{d_1 \in \mathcal{C}, d_2 \in \mathcal{C}_+ \\ f \in \mathrm{Hom}_{\mathcal{C}}(d_1 d_2, c)}} d_1 * d_2 \to c* \Big).$$

We denote by $\upsilon_c : c* \to \delta(c)$ the canonical map. It is an isomorphism if $c = \mathbf{1}$.

Lemma 3.1. *Let $u \in \mathrm{Hom}_{\mathcal{C}}(c_1 c_2, c_1' c_2')$. The composition $(c_1' \upsilon_{c_2'}) \circ (u*) \circ (c_1 \alpha_{c_2})$ vanishes if and only if $u \in L(c_1, c_2, c_1', c_2')$.*

Given $f \in \mathrm{Hom}_{\mathcal{C}}(c_1, c_1')$ and $g \in \mathrm{Hom}_{\mathcal{C}}(c_2, c_2')$, we have $c_1' \upsilon_{c_2'} \circ (h(f \otimes g)*) \circ (c_1 \alpha_{c_2}) = 0$.

Proof. Let $u = \lambda_{a,b,e}(f \otimes g \otimes h)$ with $e \in \mathcal{C}_+$. We have

$$(u*) \circ (c_1 \alpha_{c_2}) = (c_1' h \circ ge \circ c_1 f) * \circ (c_1 \alpha_{c_2}) = (c_1' h*) \circ (c_1' b \alpha_e) \circ (g * e) \circ (c_1 \alpha_a e) \circ (c_1 * f).$$

Since $(c_1' \upsilon_{c_2'}) \circ (c_1' h*) \circ (c_1' b \alpha_e) = 0$, we deduce that $(c_1' \upsilon_{c_2'}) \circ (u*) \circ (c_1 \alpha_{c_2}) = 0$.

Conversely consider $u \in \mathrm{Hom}_{\mathcal{C}}(c_1 c_2, c_1' c_2')$ such that $(c_1' \upsilon_{c_2'}) \circ (u*) \circ (c_1 \alpha_{c_2}) = 0$. There are $d_i \in \mathcal{C}$, $e_i \in \mathcal{C}_+$, $h_i \in \mathrm{Hom}_{\mathcal{C}}(d_i e_i, c_2')$, $k_i \in \mathrm{Hom}_{\delta\mathcal{C}}(c_1 * c_2, c_1' d_i * e_i)$ such that $(u*) \circ (c_1 \alpha_{c_2}) = \sum_i (c_1' h_i*) \circ (c_1' d_i \alpha_{e_i}) \circ k_i$. There is a surjective map

$$\bigoplus_{a \in \mathcal{C}} \mathrm{Hom}_{\mathcal{C}}(c_2, a d_2) \otimes \mathrm{Hom}_{\mathcal{C}}(c_1 a, d_1) \to \mathrm{Hom}_{\delta\mathcal{C}}(c_1 * c_2, d_1 * d_2)$$

$$(f_a \otimes g_a)_a \mapsto \sum_a (g_a * d_2) \circ (c_1 \alpha_a d_2) \circ (c_1 * f_a).$$

So, there are maps $f_{i,a} \in \mathrm{Hom}_{\mathcal{C}}(c_2, ae_i)$ and $g_{i,a} \in \mathrm{Hom}_{\mathcal{C}}(c_1 a, c_1' d_i)$ such that

$$(u*) \circ (c_1 \alpha_{c_2}) = \sum_{i,a} (c_1' h_i *) \circ (c_1' d_i \alpha_{e_i}) \circ (g_{i,a} * e_i) \circ (c_1 \alpha_a e_i) \circ (c_1 * f_{i,a}).$$

As a consequence, $u = \sum_{i,a} (c_1' h_i) \circ (g_{i,a} e_i) \circ (c_1 f_{i,a}) \in L(c_1, c_2, c_1', c_2')$.

The second statement follows from Lemma 2.3 and the first statement. □

Consider now $c, c' \in \mathcal{C}$ and $g \in \mathrm{Hom}_{\mathcal{C}}(c, c')$. Let $f \in \mathrm{Hom}_{\mathcal{C}}(d_1 d_2, c)$. We have $(g*) \circ (f*) \circ (d_1 \alpha_{d_2}) = ((g \circ f)*) \circ (d_1 \alpha_{d_2})$. It follows that $g* : c* \to c'*$ induces a map $\delta(g) : \delta(c) \to \delta(c')$.

It is immediate that given $h \in \mathrm{Hom}_{\mathcal{C}}(c', c'')$, we have $\delta(h \circ g) = \delta(h) \circ \delta(g)$.

4. Functor on maps

Let $c_0 \in \mathcal{C}$. We define a functor $\Delta_{c_0} : \delta\mathcal{C} \to \overline{\delta\mathcal{C}}$, whose value on $*$ is $\delta(c_0)$.

• Let us define Δ_{c_0} on objects. We put

$$\Delta_{c_0}(c * c') = \mathrm{coker}\Big(\sum_{\substack{d_1 \in \mathcal{C}, d_2 \in \mathcal{C}_+ \\ f \in \mathrm{Hom}_{\mathcal{C}}(d_1 d_2, c_0)}} (cf * c') \circ (c d_1 \alpha_{d_2} c') : \bigoplus_{\substack{d_1 \in \mathcal{C}, d_2 \in \mathcal{C}_+ \\ f \in \mathrm{Hom}_{\mathcal{C}}(d_1 d_2, c_0)}} c d_1 * d_2 c' \to c c_0 * c' \Big).$$

The identity of $cc_0 * c'$ induces a canonical isomorphism

$$\Delta_{c_0}(c * c') \xrightarrow{\sim} c \Delta_{c_0}(*) c' = c \delta(c_0) c'. \tag{1}$$

• We define now Δ_{c_0} on maps.

Let $d_1, c_1, c_2, c_1', c_2' \in \mathcal{C}$ and $d_2 \in \mathcal{C}_+$. Let $f \in \mathrm{Hom}_{\mathcal{C}}(d_1 d_2, c_0)$ and $f_i \in \mathrm{Hom}_{\mathcal{C}}(c_i, c_i')$ for $i \in \{1, 2\}$. There is a commutative diagram

$$\begin{array}{ccc} c_1 d_1 * d_2 c_2 & \xrightarrow{(c_1 f * c_2) \circ (c_1 d_1 \alpha_{d_2} c_2)} & c_1 c_0 * c_2 \\ \downarrow{\scriptstyle f_1 d_1 * d_2 f_2} & & \downarrow{\scriptstyle f_1 c_0 * f_2} \\ c_1' d_1 * d_2 c_2' & \xrightarrow[(c_1' f * c_2') \circ (c_1' d_1 \alpha_{d_2} c_2')]{} & c_1' c_0 * c_2' \end{array}$$

It follows that $f_1 c_0 * f_2$ induces a map $\Delta_{c_0}(f_1 * f_2) : \Delta_{c_0}(c_1 * c_2) \to \Delta_{c_0}(c_1' * c_2')$.

We have

$$(cf*) \circ (c d_1 \alpha_{d_2}) \circ (\tau_{d_1, c} * d_2) \circ (d_1 \alpha_c d_2) \circ (d_1 * \tau_{d_2, c}) =$$
$$(cf*) \circ (\tau_{d_1, c} d_2 *) \circ (d_1 c \alpha_{d_2}) \circ (d_1 \alpha_c d_2) \circ (d_1 * \tau_{d_2, c}) =$$
$$(cf*) \circ (\tau_{d_1, c} d_2 *) \circ (d_1 c \alpha_{c d_2}) \circ (d_1 * \tau_{d_2, c}) =$$
$$(cf*) \circ (\tau_{d_1, c} d_2 *) \circ (d_1 \tau_{d_2, c} *) \circ (d_1 c \alpha_{d_2 c}) =$$
$$(cf*) \circ (\tau_{d_1 d_2, c} *) \circ (d_1 d_2 \alpha_c) \circ (d_1 \alpha_{d_2} c) =$$
$$\Big((\tau_{c_0, c} *) \circ (fc*) - \big(h(f \otimes c)*\big) \Big) \circ (d_1 d_2 \alpha_c) \circ (d_1 \alpha_{d_2} c) =$$
$$(\tau_{c_0, c} *) \circ (c_0 \alpha_c) \circ (f * c) \circ (d_1 \alpha_{d_2} c) - \big(h(f \otimes c)*\big) \circ (d_1 d_2 \alpha_c) \circ (d_1 \alpha_{d_2} c).$$

We have

$$(c \upsilon_{c_0}) \circ \big(h(f \otimes c)*\big) \circ (d_1 d_2 \alpha_c) = 0$$

by Lemma 3.1. We deduce that the following diagram commutes

$$\begin{array}{ccccc} & & d_1 * d_2 c & \xrightarrow{(f*c)\circ(d_1\alpha_{d_2}c)} & c_0 * c & \xrightarrow{(\tau_{c_0c}*)\circ(c_0\alpha_c)} & cc_0* \\ & {\scriptstyle (\tau_{d_1,c}*d_2)\circ(d_1\alpha_c d_2)\circ(d_1*\tau_{d_2,c})}\downarrow & & & & & \downarrow{\scriptstyle c\upsilon_{c_0}} \\ & & cd_1 * d_2 & \xrightarrow[(cf*)\circ(cd_1\alpha_{d_2})]{} & cc_0* & \xrightarrow[c\upsilon_{c_0}]{} & c\delta(c_0) \end{array}$$

It follows that $(\tau_{c_0c}*)\circ(c_0\alpha_c) : c_0 * c \to cc_0*$ induces a map $\Delta_{c_0}(\alpha_c) : \Delta_{c_0}(*c) \to \Delta_{c_0}(c*)$.

We put $\Delta_{c_0}(c_1\alpha_c c_2) = c_1\Delta_{c_0}(\alpha_c)c_2$, where we have used the canonical isomorphisms (1).

- Let us now check the defining relations for the maps of $\delta\mathcal{C}$. Let $f \in \mathrm{Hom}_{\mathcal{C}}(c, c')$. We have

$$\begin{aligned}(fc_0*)\circ(\tau_{c_0c}*)\circ(c_0\alpha_c) &= (\tau_{c_0c'}*)\circ(c_0f*)\circ(c_0\alpha_c) - (h(c_0\otimes f)*)\circ(c_0\alpha_c)\\ &= (\tau_{c_0c'}*)\circ(c_0\alpha_{c'})\circ(c_0*f) - (h(c_0\otimes f)*)\circ(c_0\alpha_c)\end{aligned}$$

Lemma 3.1 shows that the composition

$$c_0 * c \xrightarrow{(h(c_0,f)*)\circ(c_0\alpha_c)} c'c_0* \xrightarrow{c'\upsilon_{c_0}} \Delta_{c_0}(c'*)$$

vanishes, hence we have a commutative diagram

$$\begin{array}{ccccc} c_0 * c & \xrightarrow{(\tau_{c_0c}*)\circ(c_0\alpha_c)} & cc_0* & \xrightarrow{fc_0*} & c'c_0* \\ {\scriptstyle c_0*f}\downarrow & & & & \downarrow{\scriptstyle c'\upsilon_{c_0}} \\ c_0 * c' & \xrightarrow[(\tau_{c_0c'}*)\circ(c_0\alpha_{c'})]{} & c'c_0* & \xrightarrow[c'\upsilon_{c_0}]{} & \Delta_{c_0}(c'*) \end{array}$$

We deduce that $\Delta_{c_0}(f*)\circ\Delta_{c_0}(\alpha_c) = \Delta_{c_0}(\alpha_{c'})\circ\Delta_{c_0}(*f)$.

We have

$$\begin{aligned}(c\tau_{c_0c'}*)\circ(cc_0\alpha_{c'})\circ(\tau_{c_0c}*c')\circ(c_0\alpha_c c') &= (c\tau_{c_0c'}*)\circ(\tau_{c_0c}c'*)\circ(c_0c\alpha_{c'})\circ(c_0\alpha_c c')\\ &= (\tau_{c_0,cc'}*)\circ(c_0\alpha_{cc'}),\end{aligned}$$

hence

$$(c\Delta_{c_0}(\alpha_{c'}))\circ(\Delta_{c_0}(\alpha_c)c') = \Delta_{c_0}(\alpha_{cc'}).$$

We deduce that Δ_{c_0} defined a functor $\delta\mathcal{C} \to \overline{\delta\mathcal{C}}$. The isomorphism (1) equips Δ_{c_0} with a structure of morphism of $(\mathcal{C},\mathcal{C})$-bimodules.

5. Naturality

Consider now $c_0, c_0' \in \mathcal{C}$ and $g \in \mathrm{Hom}_{\mathcal{C}}(c_0, c_0')$. We will define $\Delta_g : \Delta_{c_0} \to \Delta_{c_0'}$.

Let $f \in \mathrm{Hom}(d_1d_2, c_0)$. We have a commutative diagram

$$\begin{array}{ccc} cd_1 * d_2c' & \xrightarrow{(cf*c')\circ(cd_1\alpha_{d_2}c')} & cc_0 * c' \\ {\scriptstyle \mathrm{id}}\downarrow & & \downarrow{\scriptstyle cg*c'} \\ cd_1 * d_2c' & \xrightarrow[(c(g\circ f)*c')\circ(cd_1\alpha_{d_2}c')]{} & cc_0' * c' \end{array}$$

It follows that $cg * c' : cc_0 * c' \to cc_0' * c'$ induces a map $\Delta_g(c * c') : \Delta_{c_0}(c * c') \to \Delta_{c_0'}(c * c')$.

It is immediate that given $h \in \mathrm{Hom}_{\mathcal{C}}(c_0', c_0'')$, we have $\Delta_{h\circ g}(c * c') = \Delta_h(c * c')\circ\Delta_g(c * c')$.

Consider now $h \in \mathrm{Hom}_{\mathcal{C}}(c_1, c_2)$ and $h' \in \mathrm{Hom}_{\mathcal{C}}(c_1', c_2')$. We have

$$(c_2 g * c_2') \circ (h c_0 * h') = (h c_0' * h') \circ (c_1 g * c_1'),$$

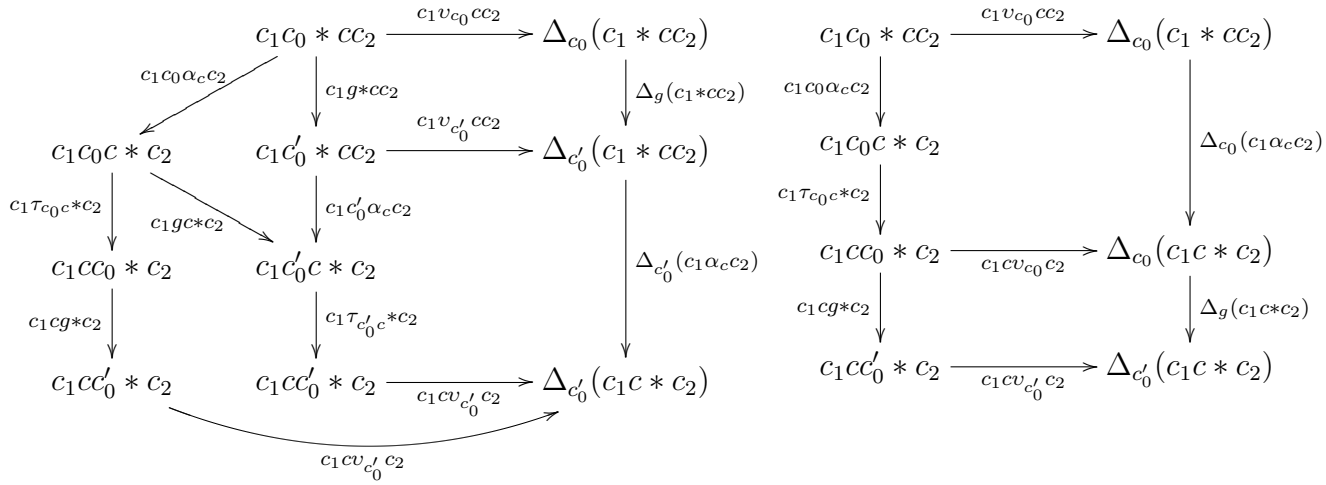
hence

$$\Delta_g(c_2 * c_2') \circ \Delta_{c_0}(h * h') = \Delta_{c_0'}(h * h') \circ \Delta_g(c_1 * c_1').$$

Let $c, c_1, c_2 \in \mathcal{C}$. Lemma 3.1 shows there is a commutative diagram

$$\begin{array}{ccccc}
c_1c_0 * cc_2 & \xrightarrow{c_1c_0\alpha_c c_2} & c_1c_0c * c_2 & \xrightarrow{c_1(\tau_{c_0'c}\circ(gc))*c_2} & c_1cc_0' * c_2 \\
\downarrow{\scriptstyle c_1c_0\alpha_c c_2} & & & & \downarrow{\scriptstyle c_1c\upsilon_{c_0'}c_2} \\
c_1c_0c * c_2 & \xrightarrow[c_1(cg\circ\tau_{c_0c})*c_2]{} & c_1cc_0' * c_2 & \xrightarrow[c_1c\upsilon_{c_0'}c_2]{} & \Delta_{c_0'}(c_1c * c_2).
\end{array}$$

The commutative diagrams



show that

$$\Delta_g(c_1c * c_2) \circ \Delta_{c_0}(c_1\alpha_c c_2) = \Delta_{c_0'}(c_1\alpha_c c_2) \circ \Delta_g(c_1 * cc_2).$$

So Δ_g is a natural transformation of $(\mathcal{C}, \mathcal{C})$-bimodule functors.

We have constructed a functor with the required properties, the non-vanishing property following from Lemmas 3.1 and 2.4.

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